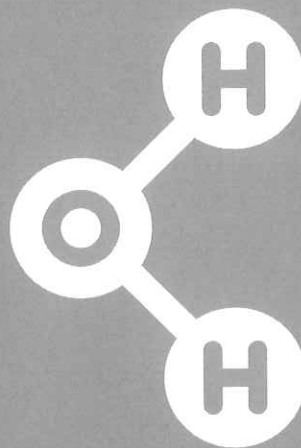




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER C



2012 ICAS

International Competitions
and Assessments for Schools

SCIENCE

**Educational
Assessment
Australia**

eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

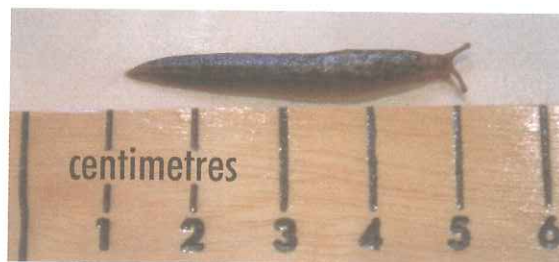
There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

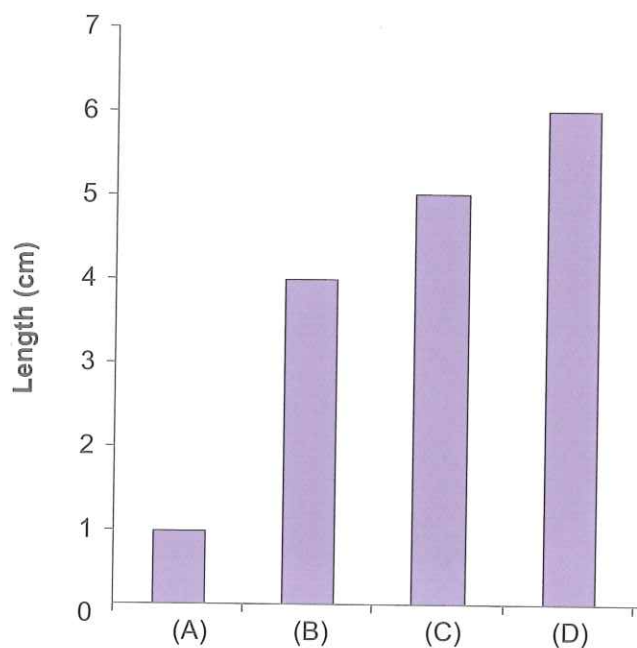
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Luca measured the length of a slug using a centimetre ruler.



Which column on the graph represents the length of the slug?



2. We hear sound when energy is transferred from the source of the sound to our ears. Loudness is measured in decibels (dB).

The table shows the loudness of different sounds.

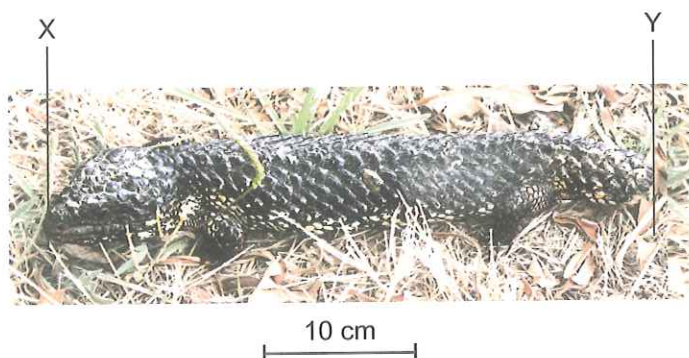
Sound	Loudness (dB)
whisper	15
raindrops	40
normal conversation	60
vacuum cleaner	70
busy road	80
hair dryer	90
rock concert	120
fire cracker	140

Sounds with a loudness greater than 100 dB can cause hearing damage if you hear them for more than 15 minutes.

Which of the following sounds could cause hearing damage?

- (A) vacuum cleaner (B) busy road (C) hair dryer (D) rock concert

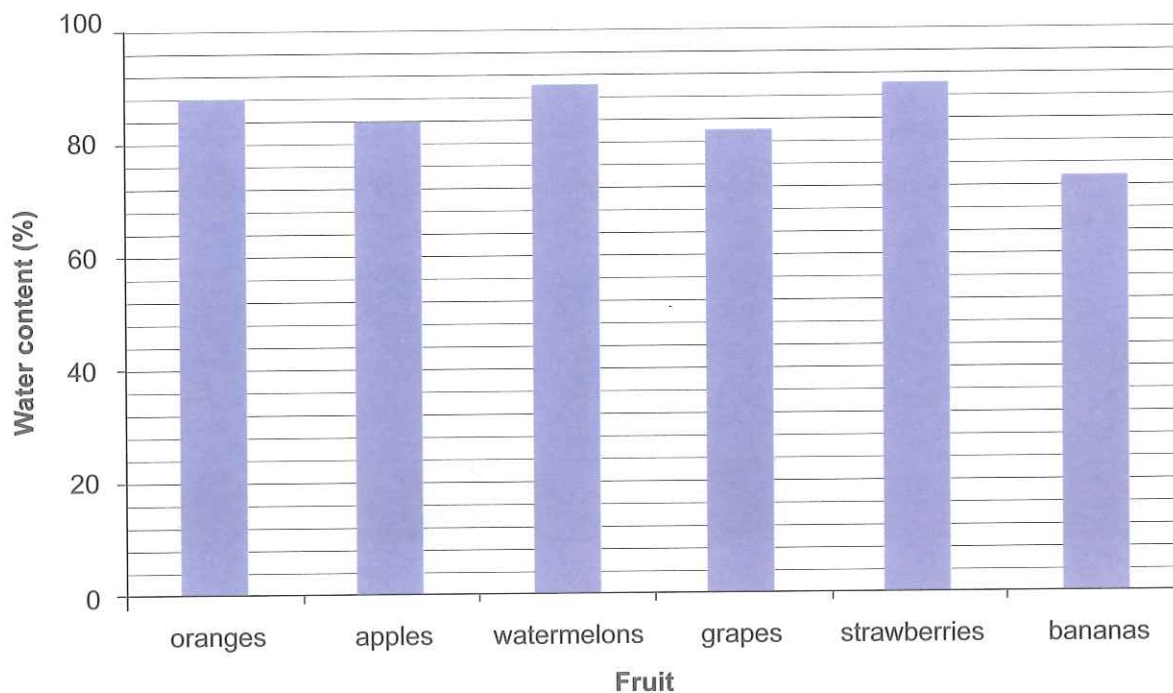
3. This photograph shows a lizard.



What is the approximate length (in cm) of the lizard from X to Y?

- (A) 8 (B) 10 (C) 20 (D) 40

4. This graph shows what percentage of the mass of some fruit is made up of water.

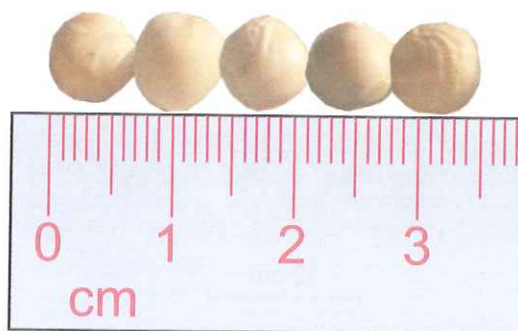


What conclusion can be drawn from this graph?

- (A) Apples contain a higher percentage of water than the other fruit.
(B) Oranges contain a higher percentage of water than strawberries.
(C) Watermelons and strawberries contain the same percentage of water.
(D) Grapes and apples contain the lowest percentage of water of these fruit.

For questions 5 and 6 use the information below.

The diagram shows some peas next to a ruler marked in centimetres and millimetres.



5. What is the approximate diameter (in millimetres) of one pea?
- (A) 5 (B) 7 (C) 10 (D) 35
6. What is the best way of getting a more accurate estimate of the average diameter of peas?
- (A) measure only the smooth peas
(B) measure the five peas one at a time
(C) measure five peas all lined up the same way
(D) measure 40 peas lined up as in the diagram

7. The photographs show a plant as it is flowering. The photographs are **NOT** shown in the order in which they were taken.



1



2



3



4

In which order were the photographs taken?

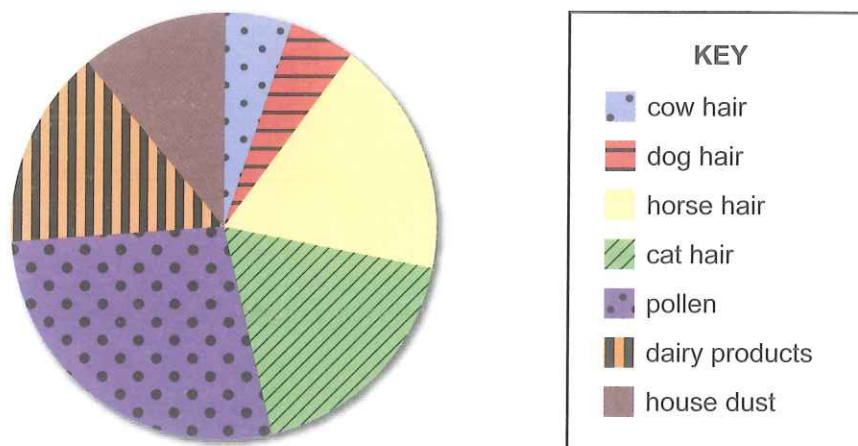
- (A) 1, 3, 2, 4 (B) 2, 3, 4, 1
(C) 3, 1, 4, 2 (D) 4, 2, 3, 1

For questions 8 and 9 use the information below.

Certain substances make some people sneeze or itch. These people have an allergic reaction to these substances.

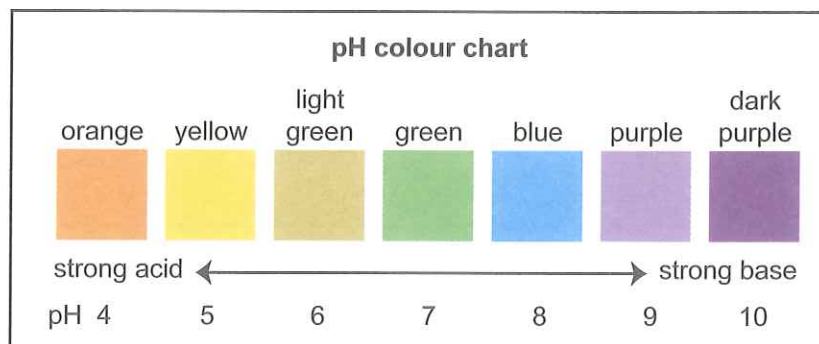
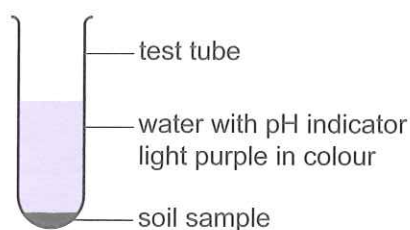
Kerry surveyed a large number of people in Sydney who suffer from allergies to find out what substances cause their allergic reactions.

The pie graph shows her results.



8. Which substance was the most common cause of allergic reactions?
- (A) dog hair (B) pollen (C) dairy products (D) house dust
9. The actual number of people who have an allergic reaction to cow hair may be much higher than the survey results show.
- What is the most likely reason for this difference?
- (A) Cow hair is very short.
 (B) People living in cities rarely come into contact with cow hair.
 (C) It is hard to tell if the reaction is caused by the cow hair or another substance.
 (D) Tests for allergic reactions to cow hair are not as accurate as they are for other substances.

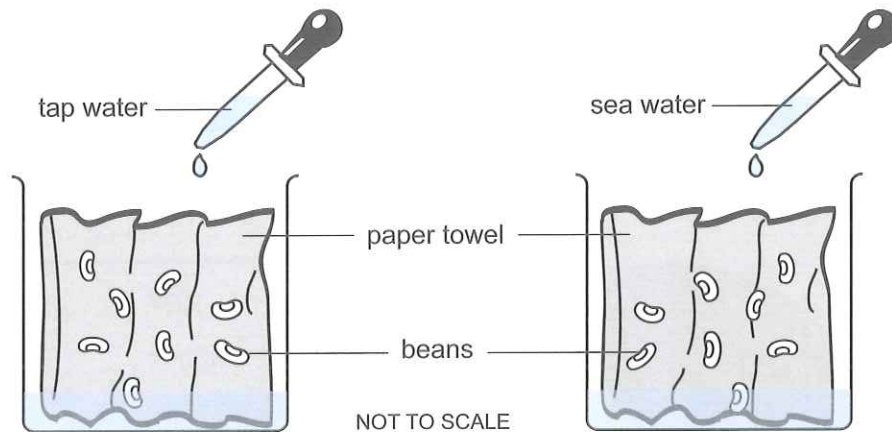
10. Tim wanted to test the pH of some soil. He mixed the soil with water in a test tube and added five drops of indicator from his pH testing kit. He matched the colour produced in the test tube against the chart supplied with the kit.



What is the approximate pH value of Tim's soil sample?

- (A) 4.5 (B) 6.5 (C) 8.5 (D) 10.5

11. Sarah saw this diagram of an experiment in a science book.

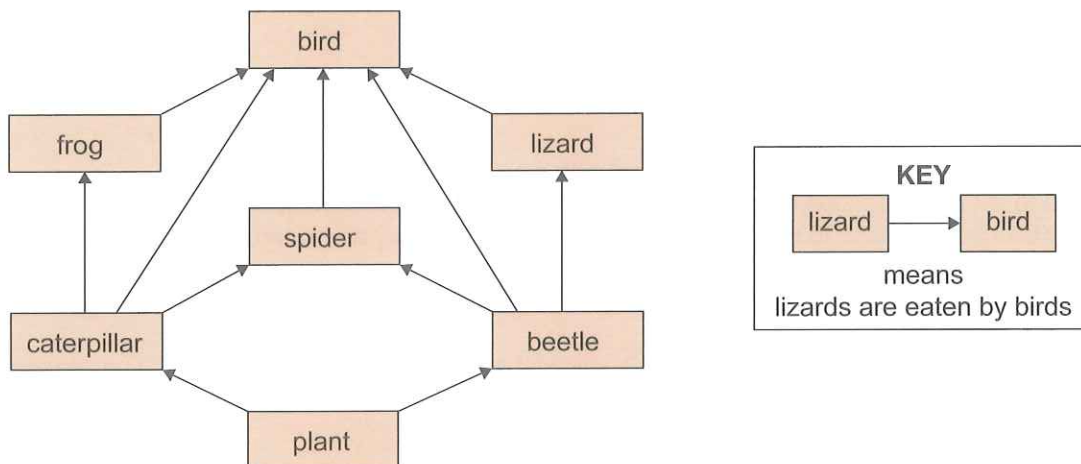


She decided to carry out the experiment herself.

Which of these questions will Sarah be able to answer from the results of her experiment?

- (A) Do beans germinate faster in groups of eight?
- (B) How much water is needed for beans to germinate?
- (C) Does a paper towel affect the germination of beans?
- (D) Does the type of water affect the germination of beans?

12. The diagram shows what some animals eat.

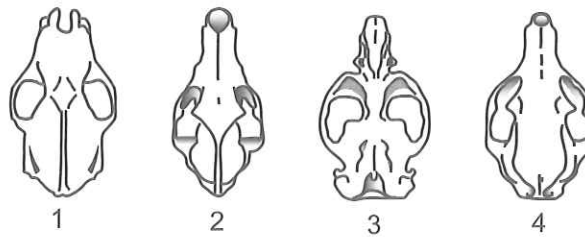


A disease killed all of the birds in this food web. Which of the following animal populations is most likely to decrease in number?

- (A) frog
- (B) lizard
- (C) beetle
- (D) spider

13. Scientists use similarities in the structure of skulls to determine how closely different animals are related.

The diagram shows the skulls of four different animals, seen from above.



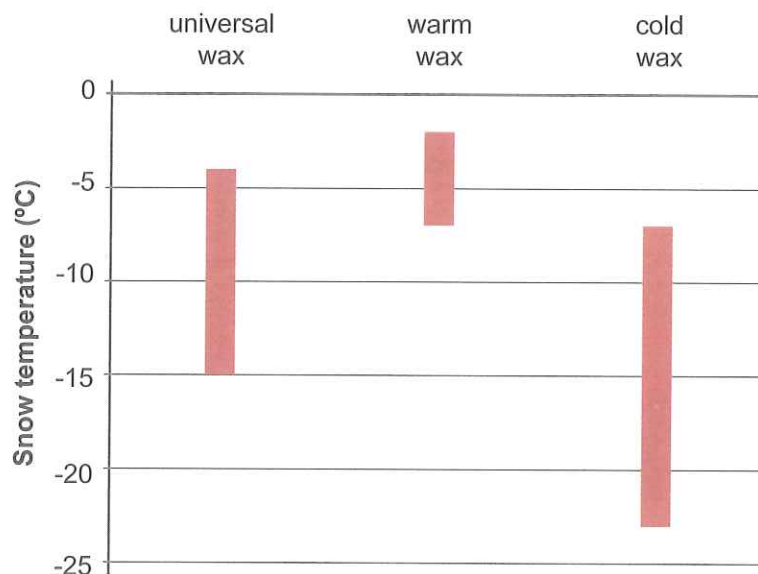
Which two animals are most closely related?

- (A) animals 1 and 4
(B) animals 1 and 3
(C) animals 2 and 3
(D) animals 2 and 4

For questions 14 and 15 use the information below.

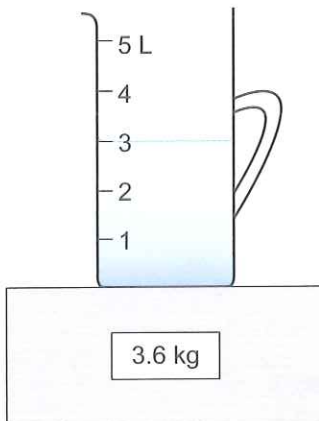
Snowboarders rub wax on the bottom of their snowboards to protect them and to improve their performance. Ice crystals can damage a snowboard by scratching it. Oxygen in the snow can cause rust to form in the metal parts of a snowboard.

The graph shows the temperature range in which each wax was designed to be used.



14. Which type of wax should be used when the snow temperature is -10°C ?
- (A) cold wax only
(B) warm wax only
(C) either universal wax or cold wax
(D) either universal wax or warm wax
15. Which statement is correct?
- (A) The wax has to be softer than the snow to absorb the oxygen from the air.
(B) The wax has to be softer than the snow to absorb scratches from the snow.
(C) The wax has to be harder than the snowboard to protect the snowboard from rust.
(D) The wax has to be harder than the snowboard to protect the snowboard from scratches.

16. Wendy measured the mass of a plastic jug containing water. The mass of one litre (L) of water is one kilogram (kg).

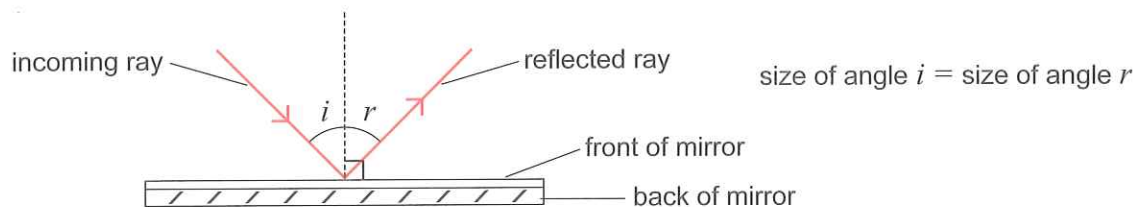


The diagram shows the mass of the jug including the water it contained.

What is the approximate mass of the jug (in kg)?

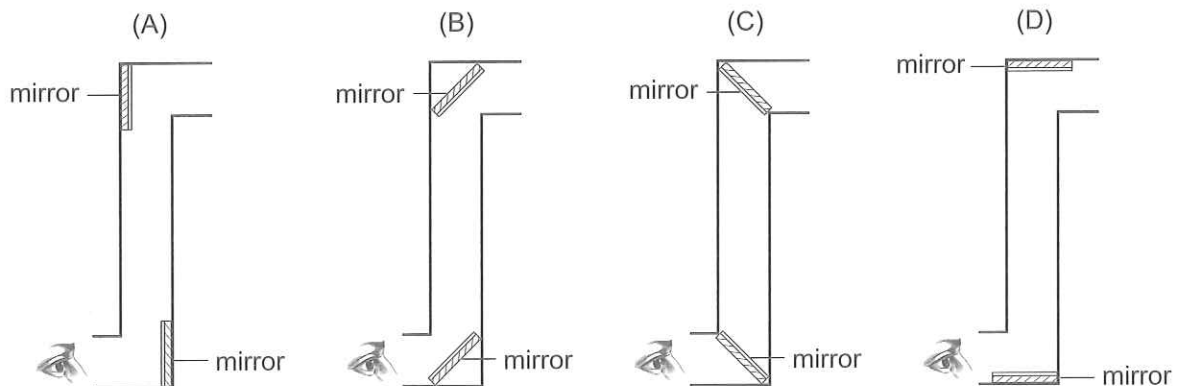
- (A) 0.6
- (B) 2.0
- (C) 2.6
- (D) 3.0

17. Light travels in straight lines. This diagram shows what happens when a ray of light hits a mirror.



A periscope enables you to look over fences or obstacles. A model periscope can be built using two mirrors inside a tube.

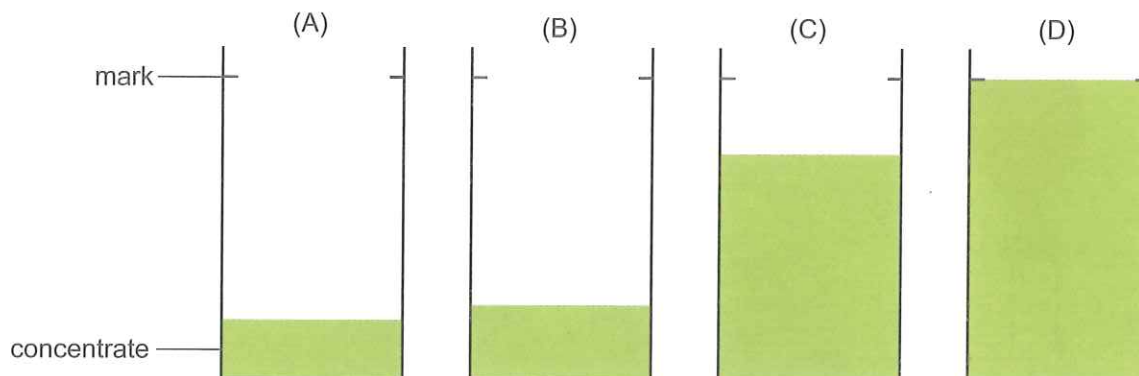
Which diagram shows the placement of two mirrors inside a tube that will let you see through it?



18. Jenny bought a bottle of fruit juice concentrate. The instructions said that she should mix 1 part concentrate to 4 parts water.

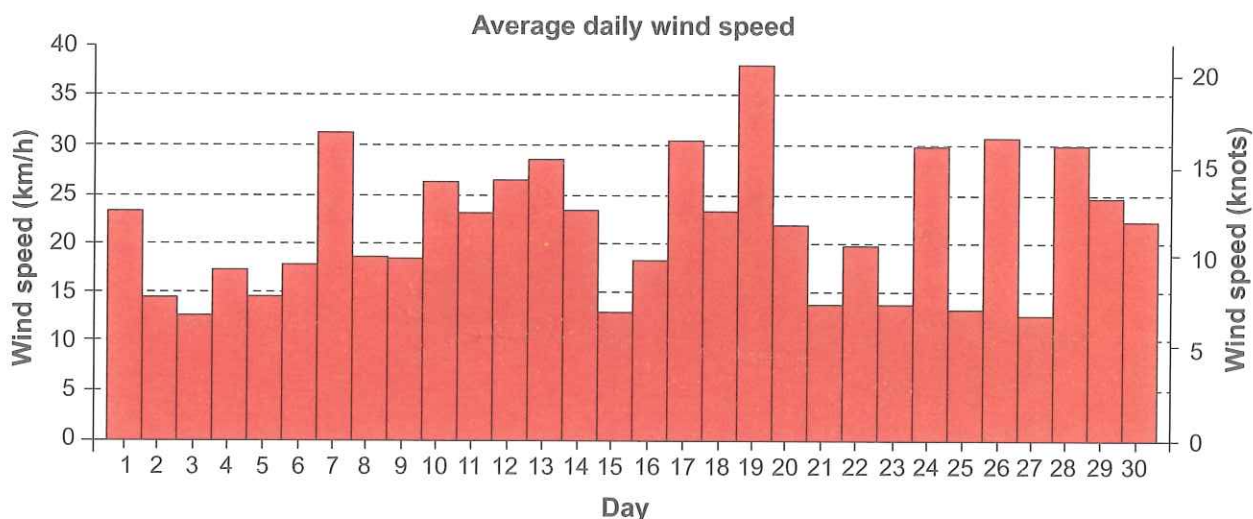
She wanted to make one container of fruit juice.

Which container shows the correct amount of concentrate that she should use, before filling to the mark with water?



For questions 19 and 20 use the information below.

This graph shows the average daily wind speed in an area over a period of 30 days.



19. What was the highest average daily wind speed, in kilometres per hour (km/h), in the 30-day period?

(A) 38 (B) 21 (C) 12 (D) 7

20. The wind speeds are given in two different units, kilometres per hour and knots.

Approximately what speed, in kilometres per hour, is equal to one knot?

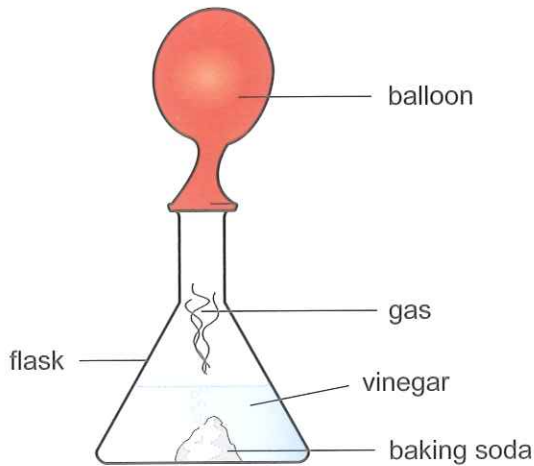
(A) 1.5 (B) 1.9 (C) 10 (D) 18

21. When vinegar is mixed with baking soda, bubbles of gas form. The gas can be trapped in a balloon.

Lina had a flask and a balloon which had a total mass of 250 g.

She put 15 g of baking soda and 55 g of vinegar into the flask and covered the mouth of the flask with the balloon before any gas could escape.

Bubbles started forming in the flask and the balloon expanded.



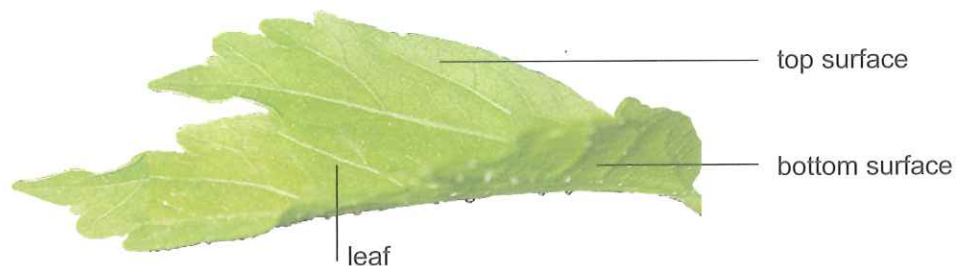
When the bubbling stopped, Lina weighed the flask with the balloon still attached without letting any gas escape.

What should be the total mass?

- (A) 250 g
- (B) 305 g
- (C) 320 g
- (D) more than 320 g

22. Leaves that have just been picked from healthy plants contain water. After a leaf is pulled off a plant, it loses water and dries up.

Water cannot escape from a leaf surface that has been coated with grease.



Bert carried out an investigation using four similar leaves taken from the same healthy plant.

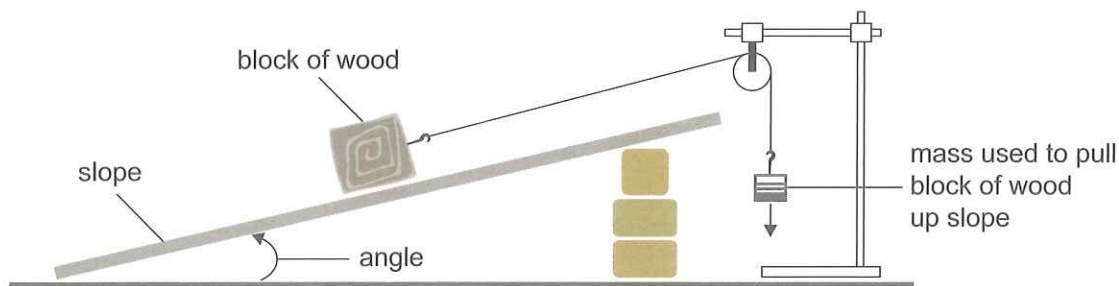
He covered some surfaces of the leaves with grease:

- Leaf 1—all of its top surface
- Leaf 2—all of its bottom surface
- Leaf 3—all of its top and bottom surfaces
- Leaf 4—no grease on either surface.

Which question is Bert trying to answer with his investigation?

- (A) Does grease enter a leaf through its top surface?
- (B) From which surface does a leaf lose more water?
- (C) How quickly does a leaf with no grease on it dry up?
- (D) Do larger leaves lose more water than smaller leaves?

23. Some students were studying friction. They set up the apparatus below to test how the surface and angle of a slope affect the movement of a block of wood.



In each trial, the students changed the angle of the slope and/or the surface, and adjusted the size of the mass until the block of wood just started to move up the slope.

The table shows the surface and angle of the slope in each trial.

Trial	Surface	Angle
A	rough carpet	40°
B	rough carpet	30°
C	smooth polished wood	40°
D	smooth polished wood	30°

Which trial required the largest mass to pull the block of wood up the slope?

- (A) A (B) B (C) C (D) D

24. Mary found four different recipes (P, Q, R and S) for making biscuits.

The table gives the ingredients and oven temperature for each recipe.

Recipe	Plain flour (cups)	Brown sugar (cups)	Coconut (cups)	Golden syrup (tablespoons)	Oven temperature (°C)
P	1	1	1	1	180
Q	1	$\frac{3}{4}$	1	2	175
R	$1\frac{1}{4}$	1	$\frac{1}{2}$	2	180
S	1	1	1	2	175

Mary made biscuits using each of the recipes and compared her results. She tasted each biscuit and concluded that using more brown sugar made the biscuits firmer and crunchier.

Which biscuits led her to this conclusion?

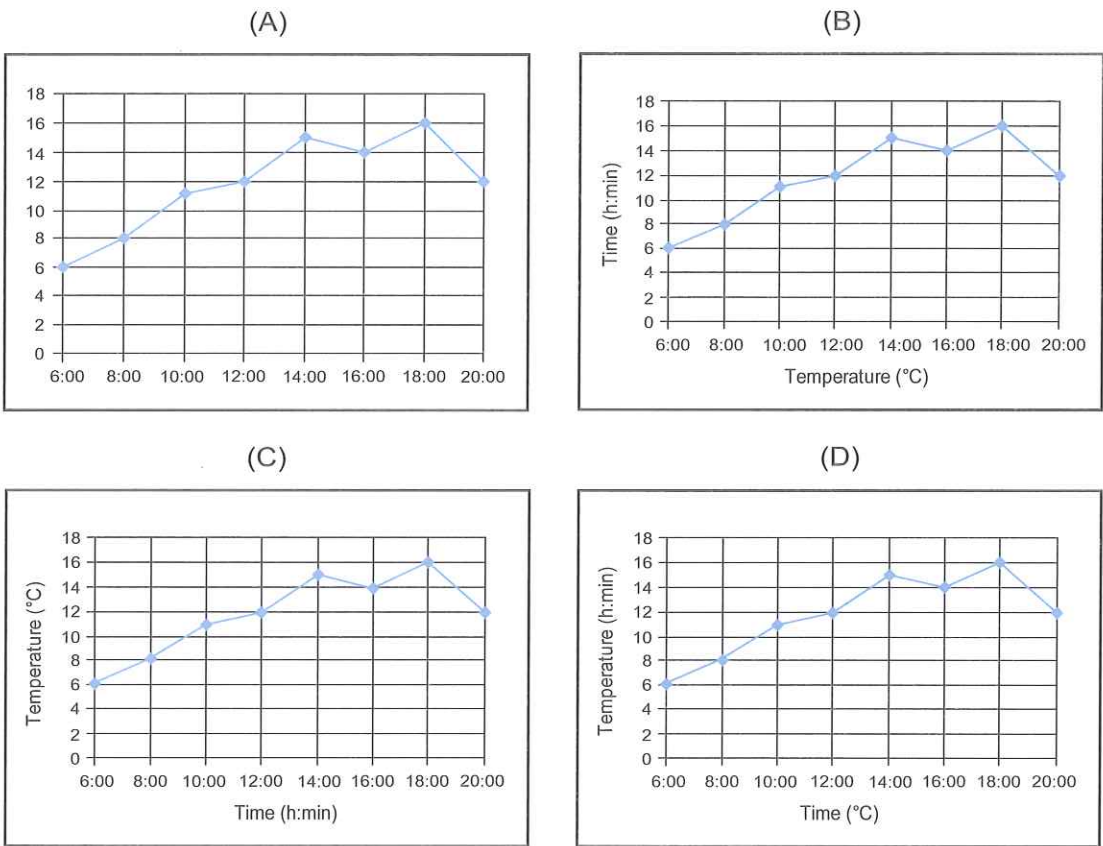
Biscuits from recipes

- (A) P and Q (B) P and R (C) Q and R (D) Q and S

25. Ann measured the air temperature at different times during the day. The table shows the data that she recorded.

Time (h:min)	Temperature (°C)
6:00	6
8:00	8
10:00	11
12:00	12
14:00	15
16:00	14
18:00	16
20:00	12

Which graph correctly shows the data Ann recorded?



26. Ben has four test tubes containing sugar solutions of different concentrations. The sugar concentrations are 5%, 10%, 15% and 20%.

Ben knows the following:

- test tube 4 contains the highest concentration
- test tube 1 does not have the lowest concentration
- test tube 2 contains a greater sugar concentration than test tube 1.

Which test tube contains the 15% sugar solution?

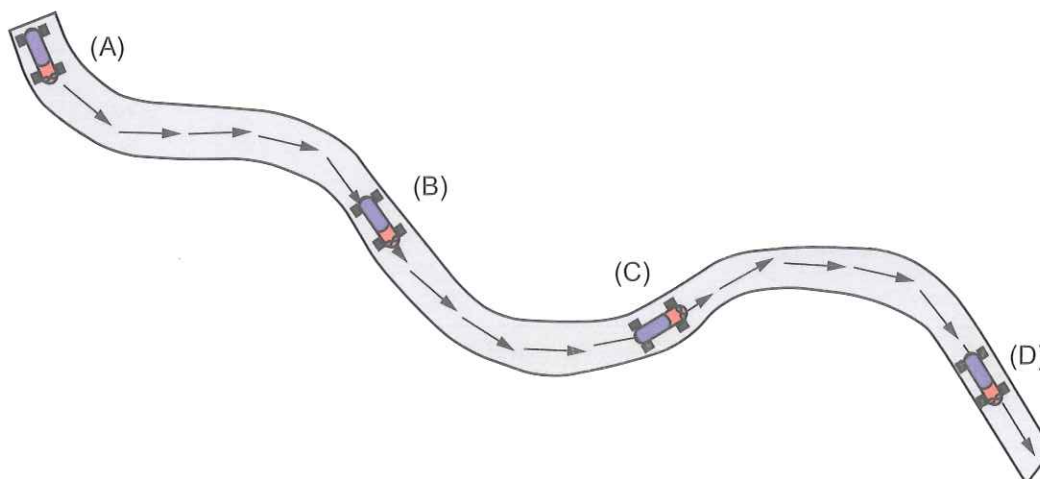
(A) 1 (B) 2 (C) 3 (D) 4

27. The photograph shows a driver's view through her front windscreen at a particular point during a journey.



The map below shows the driver's journey from (A) to (D).

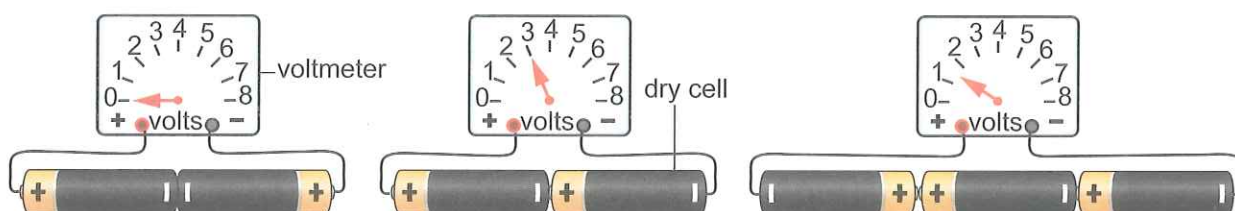
At which point along the journey would the driver see the view shown in the photograph?



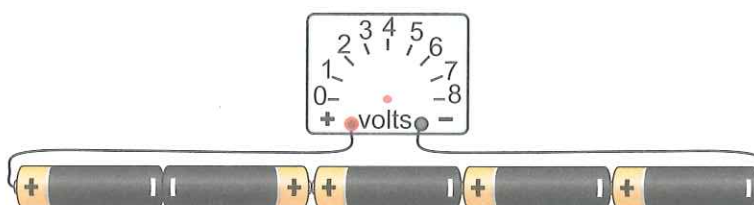
28. A dry cell produces a voltage. This can be measured using a voltmeter.

When dry cells are joined, the total voltage depends on the number of cells and how they are connected.

The diagrams show three ways that dry cells may be connected and the number of volts produced in each case.



The diagram shows some dry cells and a voltmeter connected together.



What will be the reading on the voltmeter (in volts)?

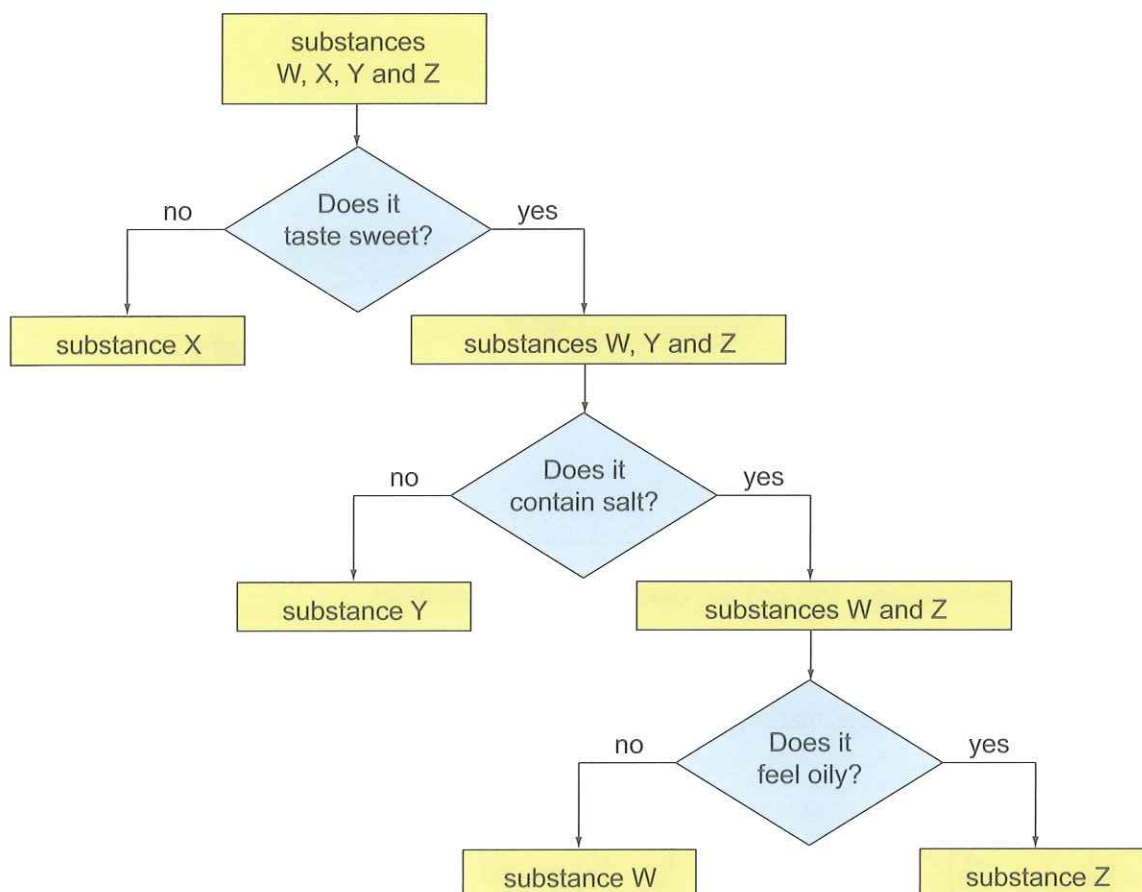
- (A) 3 (B) $4\frac{1}{2}$ (C) 6 (D) $7\frac{1}{2}$

For questions 29 and 30 use the information below.

The table lists information about four substances found in a kitchen.

Substance	Ingredients
chicken stock	salt, flour, sugar, yeast extract
drinking chocolate	sugar, cocoa, milk solids, malt extract
peanut butter	peanuts, canola oil, sugar, salt
vegemite	yeast extract, salt, malt extract, colouring

Jack used some tests as shown to distinguish between the four substances.



29. Which of the following is substance Y?

- | | |
|-------------------|------------------------|
| (A) chicken stock | (B) drinking chocolate |
| (C) peanut butter | (D) vegemite |

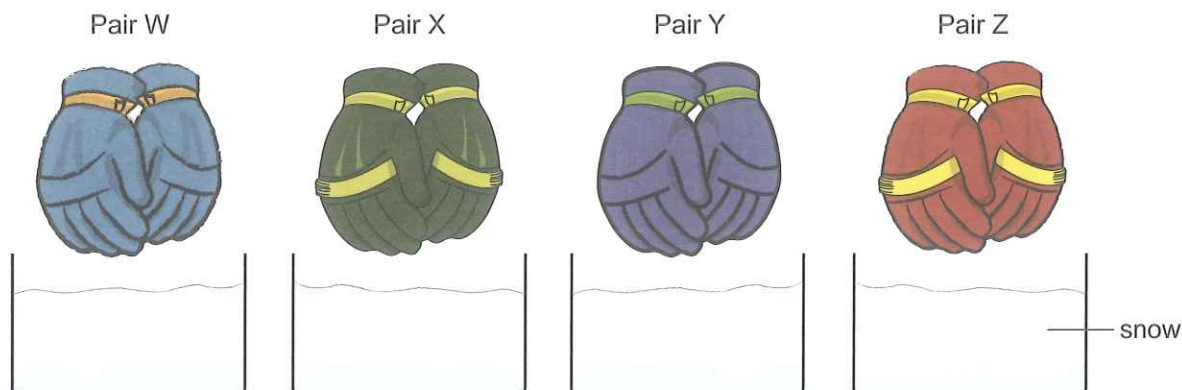
30. We use our senses when we make observations.

Which of the senses was **NOT** needed to tell substances W, X, Y and Z apart?

- | | | | |
|-----------|-------------|-----------|-----------|
| (A) sight | (B) hearing | (C) taste | (D) touch |
|-----------|-------------|-----------|-----------|

For questions 31 and 32 use the information below.

Joe carried out an investigation to see whether the material of which gloves are made affects how much water they absorb. He measured the mass of each pair of gloves before placing them in a container of snow. He measured the mass of each pair of gloves again after taking them out of the snow.



31. Which of the following did Joe have to change in his investigation?

- (A) the material of the gloves
- (B) the amount of snow in the containers
- (C) the temperature of the snow in the containers
- (D) the length of time the gloves were in the snow

32. The table shows the results of Joe's investigation.

Pair	Mass of gloves	
	Before being placed in snow (g)	After being taken out of snow (g)
W	40.0	40.1
X	60.8	60.8
Y	35.5	34.9
Z	28.8	38.7

Joe realised that he had made a mistake during the investigation.

With which pair of gloves did Joe make the mistake, and what is most likely to have made him aware of his mistake?

	He made the mistake with pair	He realised his mistake because
(A)	X.	the mass of the gloves did not change.
(B)	X.	the mass of the gloves was too great.
(C)	Y.	the mass of the gloves decreased.
(D)	Y.	the mass of the gloves changed too much.

33. Shanta considered cockroaches a pest. She read on a website that cockroaches prefer particular environmental conditions of temperature, moisture and light.

She set up eight environments to investigate which conditions attract the greatest number of cockroaches.

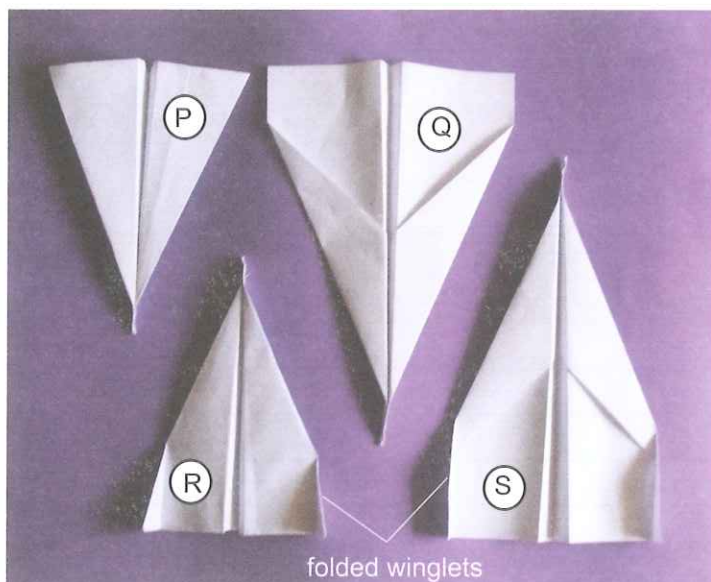


cool dry light	cool dry dark	X	cool moist dark
warm dry light	Y	warm moist dark	warm moist light

To make a fair test, what conditions did Shanta need to set for environments X and Y?

	Conditions for environment X			Conditions for environment Y		
(A)	cool	dry	light	warm	dry	light
(B)	cool	dry	dark	warm	moist	light
(C)	cool	moist	dark	warm	moist	dark
(D)	cool	moist	light	warm	dry	dark

34. Jason made four paper aeroplanes. The photograph shows Jason's aeroplanes.

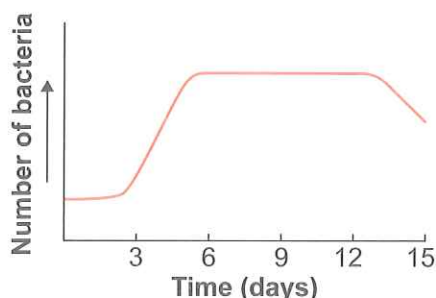


He wanted to find out which aeroplane will glide the furthest.

The results for which pair of aeroplanes will provide a fair test of how the folded winglets affect the distance an aeroplane will glide?

- (A) P and Q
(B) Q and R
(C) Q and S
(D) R and S

35. The graph shows the change in the number of bacteria living on a slice of stale bread over a given time.

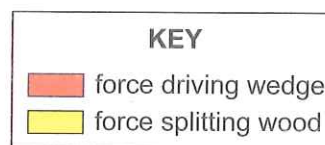
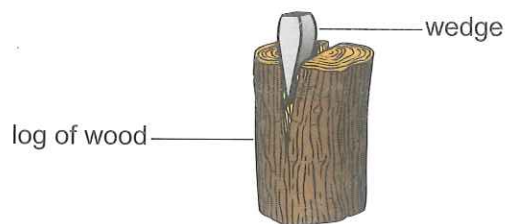


On which days was the number of bacteria that died greater than the number of bacteria that formed?

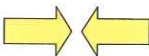
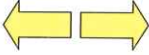
- (A) days 1–3 (B) days 3–6 (C) days 6–12 (D) days 13–15

36. A wedge is a simple machine that is used to change the direction of a force. It can be used to split a log of wood.

The wedge is driven into the wood by hitting it with a hammer.



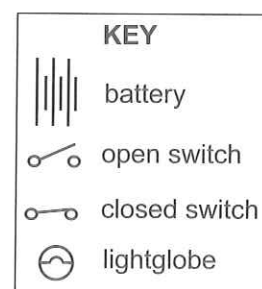
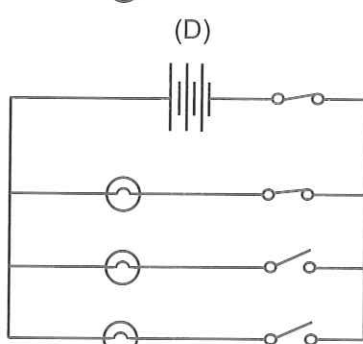
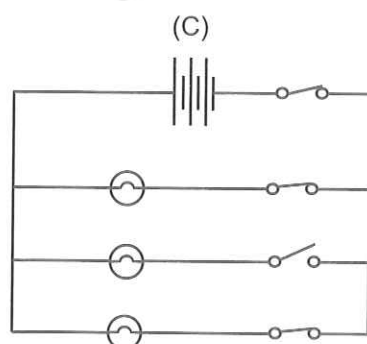
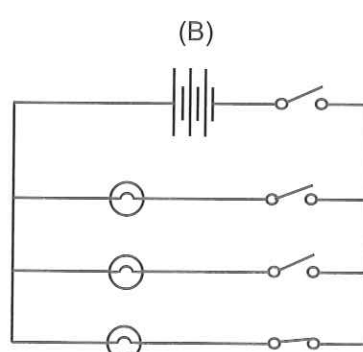
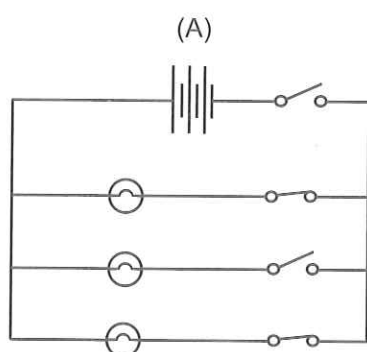
Which of the following shows the direction of the force driving the wedge and the direction of the force splitting the wood?

	Force driving wedge	Force splitting wood
(A)		
(B)		
(C)		
(D)		

37. A circuit diagram shows how an electrical circuit is joined together. Standard symbols are used in circuit diagrams to represent electrical components.

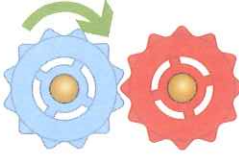
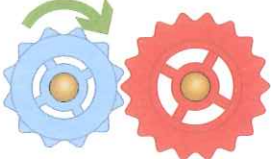
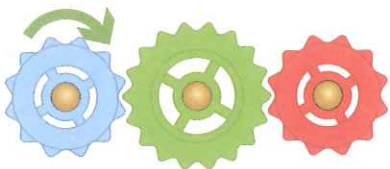
Electrical current will flow only when a circuit is closed. Lightglobes will glow only if electrical current is flowing.

The diagrams show four circuit diagrams. In which circuit would only one lightglobe glow?

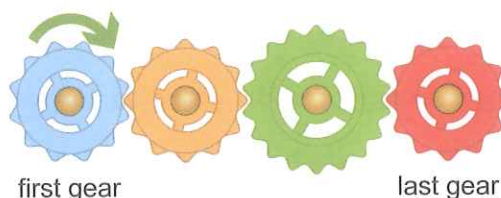


38. Gears are wheels with teeth.

The table shows three sets of gears. In each set, the first gear makes 30 turns per minute in the direction shown. For each set of gears, the table shows the direction of rotation and speed of rotation of the last gear.

	Last gear	
	Direction of rotation	Speed of rotation
 first gear last gear		30 turns per minute
 first gear last gear		20 turns per minute
 first gear last gear		30 turns per minute

This diagram shows another set of gears. The first gear makes 30 turns per minute in the direction shown.

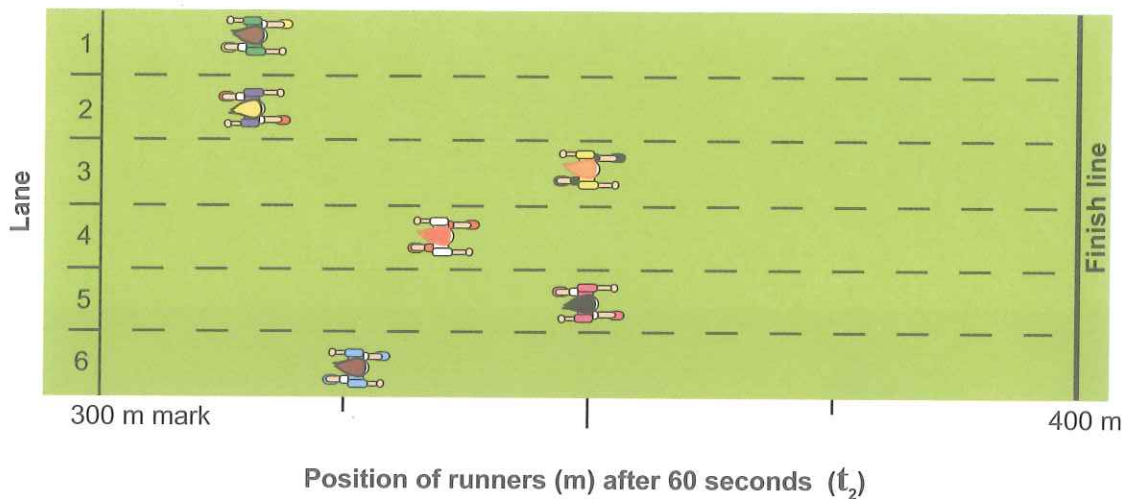
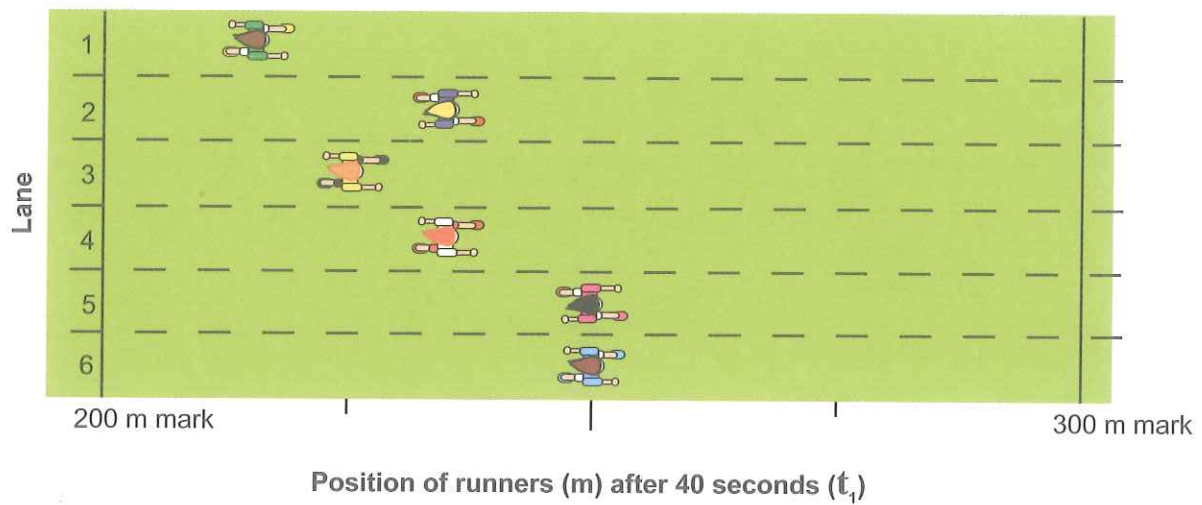


For this set of gears, what are the direction and the speed of rotation of the last gear?

	Direction of rotation	Speed of rotation
(A)		20 turns per minute
(B)		30 turns per minute
(C)		20 turns per minute
(D)		30 turns per minute

For questions 39 and 40 use the information below.

Six children competed in a 400 metre (m) race. The diagrams show the position of each child at two different times (t_1 and t_2) during the race.



39. How many metres did the child in Lane 2 run between time t_1 and time t_2 ?
- (A) 80 (B) 90 (C) 100 (D) 320

40. The speed of a moving person or object is given by the equation

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

At what speed (in metres per second) did the child in Lane 1 run between time t_1 and time t_2 ?

- (A) 5 (B) 10 (C) 15 (D) 16

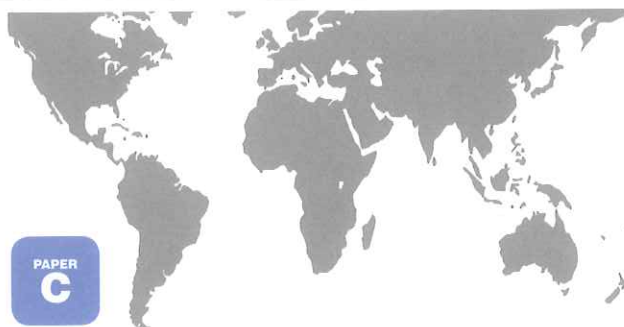
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The following year levels should sit THIS Paper:

Australia	Year 5
Brunei	Primary 5
Indonesia	Year 6
Malaysia	Standard 5
New Zealand	Year 6
Pacific	Year 5
Singapore	Primary 4
South Africa	Grade 5



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	2012
1	B
2	D
3	D
4	C
5	B
6	D
7	C
8	B
9	B
10	C
11	D
12	D
13	D
14	C
15	D
16	A
17	B
18	A
19	A
20	B
21	A
22	B
23	B
24	D
25	C
26	B
27	C
28	B
29	B
30	B
31	A
32	C
33	D
34	C
35	D
36	B
37	D
38	D
39	A
40	A